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Commuter characteristics in mass rapid transit stations in Singapore

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ABSTRACT

A study on building occupant characteristics in mass rapid transit (MRT) stations in Singapore is reported in this paper. The characteristics studied include age group and gender distributions, commuters' response to fire emergency alarms and their awareness of locations of exits, their walking speeds on walkways and stairs. The study took advantage of the fare card system used in Singapore to ascertain the age group spectrum of the commuters. Surveys were conducted among the MRT commuters to establish their intended response to three types of fire alarm systems, namely the fire alarm bell, the live announcement and the pre-recorded message. Commuter walking speeds were measured with the aid of video recording and on-site observation. The distributions of age groups in the MRT commuters were found to be statistically significantly different from that of general public, as there were mainly adults and fewer children and elderly in comparison to the general population. The distributions of gender groups in the MRT commuters were also found to be statistically significantly different from that of general public, as the female/male ratio among the commuters was greater than that in the general population. The survey results indicated that commuters intended more likely to respond to live announcement warning system than to the other two systems. The commuters' awareness of egress routes was poor. It was also noted that there was no statistically significant difference in the responses by the two gender groups to the three fire alarm warning systems but the study indicated the statistically significant gender difference in the awareness of fire escape routes. A significantly large proportion of commuters was found unaware of or unfamiliar with the current fire escape route provisions in the MRT stations from the survey. The unimpeded walking speed was found to be gender and age dependant and it may not be appropriate to apply one walking speed for all commuters. Weak correlations were found for linear fit of walking speeds over a wide range of pedestrian density on walkways and staircases.

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1. Introduction

1.1. Background

Constructions of mass rapid transit (MRT) systems are increasing worldwide with the needs to provide efficient public transport systems. In the event of fire emergency, there is potential of significant numbers of death and/or casualties. Several major MRT station fires had taken place. In February 2003, 193 were killed in an arson attack at a train leading to a MRT station fire in the South Korea City of Daegu [1]. The escalator and ticket hall fire in King's Cross Station of London Metro caused 31 deaths and 100 injured in 1987 [2].

With the implementation of performance-based approach to fire safety design, engineering principles are allowed in fire safety design to achieve the objectives and satisfy the performance requirements of building codes. Fire engineering assessment would involve identification of occupant characteristics in buildings. It is stated in the International Fire Engineering Guidelines [3] that understanding the likely nature of the building's occupants is an important element. The essential information on MRT commuters required for fire engineering assessment is as follows:

(a) Distribution—age, gender, etc.
(b) Mental attributes—level of understanding, potential emergency behaviour and familiarity with the building occupancy.
(c) Physical attributes—walking speed, mobility, etc.

Commuters' level of interpretation to fire warning/alarm signal is critical because fire alarm would initiate subsequent reactions and activities in most instances. Benthorn and Frantzich [4] stated that "certain signals are interpreted differently in different buildings." They also stated that "The knowledge of which exit

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Nomenclature

c	number of columns of a data table
D	degree of freedom
F	pedestrian flow volume (person/m min)
f_e	expected frequencies
f_o	observed frequencies
N	sample size
r	number of rows of a data table
r^2	correlation coefficient
s	standard deviation of sample results
v	walking speed (m/s)
X	sample mean
V	Cramer's V
γ	space per pedestrian (m^2/person)
ρ	pedestrian density (person/m^2)
μ	commuter population mean
μ_0	general population mean
σ	standard deviation

route that will be chosen and used in an evacuation situation is far more important in the planning stage of the evacuation system than the exit widths or lengths." Thus, commuters' awareness of exit locations should be part of the considerations for their mental attributes and familiarity with the building when the occupant characteristics are evaluated.

Walking speed is the main consideration for emergency movement [5,6] and it varies with pedestrian density and individuals [6–8]. Shao and Murosaki [8] conducted studies in Osaka and Taipei. They highlighted that "due to the difference that comes from countries, culture, social background, etc., the regional characteristics of urban evacuation will appear."

1.2. Objectives of the current study

Singapore has a well-established MRT system with more than 60 stations. Although there have been studies on pedestrian characteristics [7,8,12] and information on the local public is available [13], there is lack of knowledge on the Singapore MRT commuter characteristics in terms of age distribution, interpretation of fire warning system, familiarity to the building, choice of exit and walking speeds. It may not be appropriate to assume that the commuter population has the same characteristics as the public or to assume that data obtained from other parts of the world can be readily used to characterise the local pedestrians in the design and evaluation of particular building occupancies. The objectives of the research project reported herein are to understand above-mentioned characteristics of commuters, to identify the differences between the commuter characteristics and that of general population, and to identify the differences between the local commuter characteristics and that published elsewhere in the literature. The findings from this research can be used for design, planning and evaluation of emergency management.

The more detailed objectives of this research project are as follows:

(a) To derive the occupant distributions in terms of age and gender groups in MRT stations.
(b) To find out commuters' likely response to different types of fire alarm warning systems and their awareness of fire emergency evacuation route.
(c) To measure the walking speeds of people on walkway and stairs.

2. Methodology

2.1. Establishment of occupant groups

The occupants of MRT stations, or the commuters, were grouped according to their age and gender. The gender distributions of the commuters were obtained from on-site observations, or observations conducted at various MRT stations, during morning and evening peak hours on weekdays.

The age group distribution was obtained from the Ez-link card system, which is the public transport fare card system used in Singapore. The distributions of different age groups were from fare card transaction data recorded in the Singapore MRT systems. There are different types of public transport fare cards used by different groups of people. Besides a type of single trip fare card, there are different types of stored value fare cards catered for students at various levels of education, pre-school children, adults, senior citizens, full time national service men, staff of public transport operators, etc.

The scope of the study was streamlined to the proportions of commuters in the categories of children, elderly and adults. The commuters who hold children cards and primary school cards are in the age group of 2.5–12 years old. People who are more than 65 years old are eligible to get an elderly pass and they are classified under the elderly group in the study. The data were collected in all 63 operational stations on a typical weekday in 2005. The data registered the number of commuters checking out of the stations for the whole day.

The data collected from the MRT system were then subjected to statistical analysis and hypothesis tests to check whether there was any statistically significant difference between the commuter age distributions at MRT stations and that of the general population in Singapore.

2.2. Survey

Surveys are often used to measure facts (post-behavioural experience), attitudes (or preferences) or beliefs (assessment of importance and predictions) [19]. It is particularly useful when there is requirement to collect data on aspects of behaviour or attitudes or belief that are not easy to observe directly [20]. The survey conducted in the current study provided accounts of commuters' attitudes towards hypothetical fire incidents.

Surveys were conducted at the exits of MRT stations to determine the commuters' awareness of exit locations and how they would respond to different types of fire alarm warning systems and their familiarity to fire emergency evacuation route. A questionnaire was developed to obtain the commuters' response to three types of warning systems, namely, fire alarm bells, live announcement and pre-recorded messages. The responses for the intended actions were divided into four categories: ignore, observe others, approach staff and leave immediately [15]. The interviewees were asked to identify the fire escape routes they would take during fire. If a subject could not identify the location of any fire exit in the station, he/she was deemed unaware of the exit location. The questionnaire also included a question as to whether the subject had any experience of fire incidences in the past. The survey forms were distributed to commuters randomly.

The sample size of 400 was targeted for the surveys based on adoption of confidence level of 95%, which is usually used for most research project [9]. The sample was stratified into 189 males and 211 females. The results obtained from the survey were analysed to examine any difference between different gender groups. The survey sample was representative of gender groups but not age group because it is legally more complicated to obtain information from children. χ^2 tests were used for independence test between two categorical variables. The two variables were gender group and different responses to each question in the survey questionnaire. Besides hypothesis, test-to-test statistical significance, computation of measure of association, Cramer's *V*, was done to assess the strength of the association.

2.3. Walking speed

Video surveillance was used to record people movement at MRT facilities during morning peak hours in order to derive walking speeds under different congestion levels. Congestion level or pedestrian density is measured in terms of person/m^2 and Fruin's [7] congestion criteria were used. On-site observations were also conducted to capture unimpeded/normal speeds of people of different age and gender groups. Under congested conditions, walking speeds were not analysed against the age and gender in the current study. Walking speed was derived from the time taken by randomly identified person to cover a designated distance on walkways or staircases.

Unimpeded average walking speeds and regression lines of walking speeds on walkways, staircases were also evaluated against some guidelines or standards. Fahy and Proulx [10] compiled the walking speeds for use in evacuation modelling. They recommended 1 m/s for average walking speed under normal and unimpeded conditions.

The Singapore's code for transit system, Standard for Fire Safety in Rail Transit System (SFSRTS) [11], specifies the average walking speeds of 1, 0.3 and 0.25 m/s on walkway, down stairs and up stairs, respectively. These walking speeds were used to compare with the data obtained from this study.

Crowd movement has been quantitatively correlated in terms of speed-pedestrian density and flow volume-space per pedestrian by a number of researchers [7,8,12]. Note that the term "pedestrian density" was given different definitions in the literature. In the current paper, the pedestrian density (person/m^2) and the space per pedestrian (m^2/person) are differentiated.

Fruin's levels of service (LOS) [7] on walkways and staircases were adopted as a guide to consider average unimpeded walking speeds. For consideration of walking speed on walkway, the pedestrian density should be less than 0.43 person/m^2. On staircases, the pedestrian density should be less than 0.72 person/m^2. These pedestrian density criteria corresponded to LOS A and B on both walkways and staircases. Under LOS A and B, Fruin [7] claimed that "sufficient space is available to select normal walking speed". Fruin's [7] prediction equations on commuter walking speed-density and pedestrian flow volume-density on walkways and staircases were chosen to compare against the empirical observation data of this project. Fruin presented the results of the survey on commuters in bus terminals and transit stations. His book, *Pedestrian Planning and Design* [7], compiled the most comprehensive findings on pedestrians. Fruin [7] conducted surveys on walking speeds under different pedestrian density in central business districts of New York and Pittsburgh, though there was no clear indication of number of pedestrians involved. His model could serve as a useful reference to compare against the derived data.

3. Results and discussions: age and gender distributions

3.1. Age group distributions

Based on the statistics figures from the Ministry of Community Development, Youth and Sports (MCYS) of Singapore in 2003 [13], residents of age less than 14 years old and over 65 years old were found to be 21% and 8%, respectively, in the whole population. Despite there is no exact figure for children less than 12 years old, the percentage of children population between 2.5 and 12 years old was pro-rated proportionally to be 14.2%. Hypothesis test was used to check whether the children proportion in the age distribution of the general population was different from that obtained from the fare card counts. The *t*-test was conducted to test the hypothesis that $\mu \neq \mu_0$ for all age groups. The confidence interval of 0.025 was used. The corresponding critical value is 2.0 [14].

The test statistics for all age groups were found to fall outside of confidence interval. Therefore, the null hypotheses are rejected and the hypotheses are accepted. The commuter population percentages of all age groups were different from that of general population. A substantial proportion of children and elderly in the general population do not take MRT system or travel less frequently in MRT than adults.

3.2. Gender distributions

From observations made in all 63 MRT stations, majority of the commuters was female and male was about 6% less than female. The observations were done in the morning and evening peak hours in order to capture large crowds of commuters [15].

Based on data obtained from the Ministry of Community Development, Youth and Sports [13], the percentage of male resident population in Singapore is 49.8%, which is greater than the observed male commuter population proportion of 46.95% in MRT stations. Statistical *t*-test confirmed that the male commuter proportion is different from that of general population.

4. Results and discussion: surveys

4.1. Reactions to different types of fire warning systems

The information collected in the survey were analysed to demonstrate the criticality of the type of cue received by occupants to prompt evacuation movement. It also revealed the differences in people's perceptions to the different warning or cue received.

Presented in Fig. 1 are the percentage distributions of commuters' intended actions in response to three types of fire alarms. It is shown that a larger proportion (86.5%) of people being surveyed claimed that they would intend to leave immediately if they heard live announcement made by the station staff as compared with the proportion (27.7%) when a fire alarm bell only was heard or that (76.9%) when pre-recorded voice messages were heard.

Most commuters replied that they intended not to leave immediately when they heard sounds of fire alarm bell, which could possibly be regarded as an ambiguous initial cue. It is therefore speculated that commuters may need verification and confirmation before they act or evacuate. Therefore, if definitive information relative to the fire incident could be conveyed to the commuters, more commuters would have appropriate response.

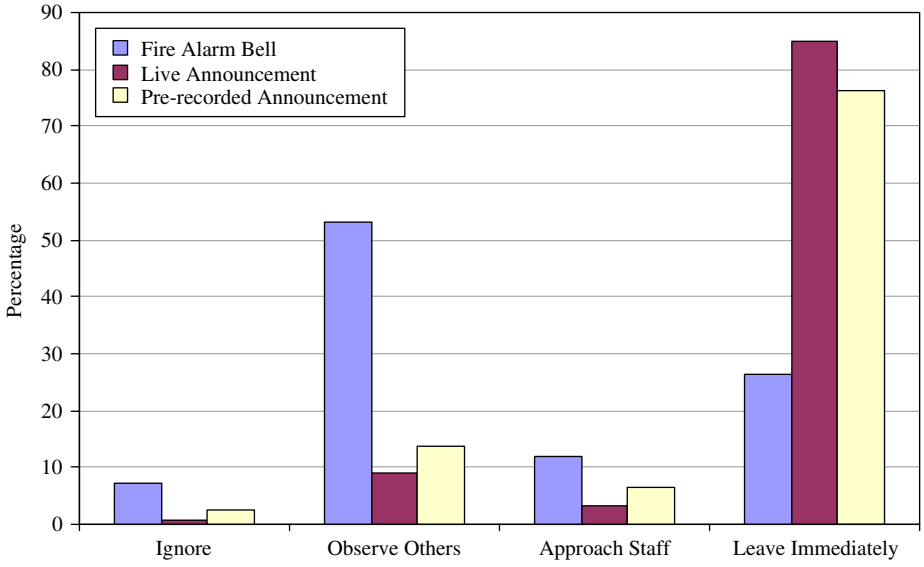


Fig. 1. Intended responses to fire alarm warning systems.

Table 1
Observed frequencies of intended reactions to fire alarm bell

Gender	Ignore	Observe others	Approach staff	Leave immediately	Subtotal (column marginal) and total
Male	16	103	20	50	189
Female	12	111	27	61	211
Subtotal (row marginal)	28	214	47	111	400

Table 2
Observed frequencies of intended reactions to live announcement

Gender	Ignore	Observe others	Approach staff	Leave immediately	Subtotal (column marginal) and total
Male	0	16	9	164	189
Female	3	22	4	182	211
Subtotal (row marginal)	3	38	13	346	400

Table 3
Observed frequencies of intended reactions to pre-recorded voice message

Gender	Ignore	Observe others	Approach staff	Leave immediately	Subtotal (column marginal) and total
Male	5	25	15	144	189
Female	5	30	12	164	211
Subtotal (row marginal)	10	55	27	308	400

The observed frequencies of intended actions, f_o, by commuters after hearing three types of alarms are listed in Tables 1–3. It can be discerned from these tables that both genders responded similarly with reception of different fire alarm warning systems. χ^2 tests were conducted to further check any dependency of the responses on gender. The null hypothesis is that the response to fire alarms is independent of gender.

In order to carry out the χ^2 test, the expected frequencies, f_e, of intended actions were calculated from the results presented in Tables 1–3 using [16]

$$f_e = (\text{row maginal} \times \text{column marginal})/\text{total number of cases} \quad (1)$$

The results are shown in Tables 4–6.

The χ^2 (obtained) values of the three types of fire warning systems were estimated according to [16]

$$\chi^2\ (\text{obtained}) = \sum \frac{(f_o - f_e)^2}{f_e} \quad (2)$$

and the results are presented in Table 7 together with the χ^2 (critical) value which was obtained from table for distribution of χ^2 in Larsen et al. [14] with the degree of freedom of 3 and significance level of 0.05. It is seen from Table 7 that all three tests fail to reject the null hypotheses since the χ^2 (critical) is higher than χ^2 (obtained) for these tests. It is, therefore, concluded that the reactions of commuters to different fire warning systems were independent of gender.

Table 4
Expected frequencies of intended reactions to fire alarm bell

Gender	Ignore	Observe others	Approach staff	Leave immediately	Total
Male	13	101	22	52	189
Female	15	113	25	59	211
Total	28	214	47	111	400

Table 5
Expected frequencies of intended reactions to live announcement

Gender	Ignore	Observe others	Approach staff	Leave immediately	Total
Male	1	18	6	163	189
Female	2	20	7	183	211
Total	3	38	13	346	400

Table 6
Expected frequencies of intended reactions to pre-recorded voice message

Gender	Ignore	Observe others	Approach staff	Leave immediately	Total
Male	5	26	13	146	189
Female	5	29	14	162	211
Total	10	55	27	308	400

Table 7
χ^2 test on fire alarm warning systems

χ^2 (obtained)			χ^2 (critical)
Alarm bell	Live announcement	Pre-recorded voice messages	
1.854	4.720	0.718	7.815

Table 8
Cramer's V test on gender dependence of intended response to fire warning systems

Alarm bell	Live announcement	Pre-recorded voice messages
0.068	0.109	0.0424

Furthermore, Cramer's V test was conducted to determine the strength of the association between the response to each type of fire alarm warning systems and gender groups. Cramer's V is a χ^2-based measure of strength of association. The formula of Cramer's V [17] is

$$V = \sqrt{\frac{\chi^2}{N \times \min(r-1, c-1)}} \tag{3}$$

where χ^2 is the obtained value from χ^2 test and N is the sample size ($= 400$). The Cramer's V test results are shown in Table 8. The maximum V value is 0.1086. This implies that the association between the response to each type of fire alarm warning systems and genders groups was weak [16].

4.2. Awareness of fire escape route

The interviewees were asked to name or describe the fire escape routes in the MRT stations. Significant proportions, namely 62.5% and 52.4%, of the female and male, respectively, could not tell the locations of the emergency exits in the MRT stations (see Fig. 2) despite these MRT stations were used by most of interviewees almost everyday. About one-third of the interviewed commuters chose their normal means into the MRT stations as the only means of fire escape routes. There were only one out of 10 of the respondents able to identify both the emergency exits at the two ends of the MRT stations and normally used staircases/escalators as means of escape routes. Most commuters did not realise that there was emergency staircase at each end of the MRT stations.

The definite conclusion in regard to the first awareness result in Fig. 2 could not be drawn with confidence level of 95% because of the relatively small stratified sample sizes (189 for male and 211 for female). However, a conclusion concerning the last awareness in this figure could be drawn. The statistical analysis showed that combined error bound (8.4%) of the estimates for two genders is less than the difference (9.1%). In other words, the hypothesis that male commuters are more likely to be able to identify the alternative exits routs than the female counterparts can be accepted, although the likelihood for either gender is small. As for the second awareness, it can be said that there is no significant difference between the two gender groups.

With reference to the observed number and the expected number of different responses presented in Tables 9 and 10, respectively, χ^2 test was conducted to verify the null hypothesis

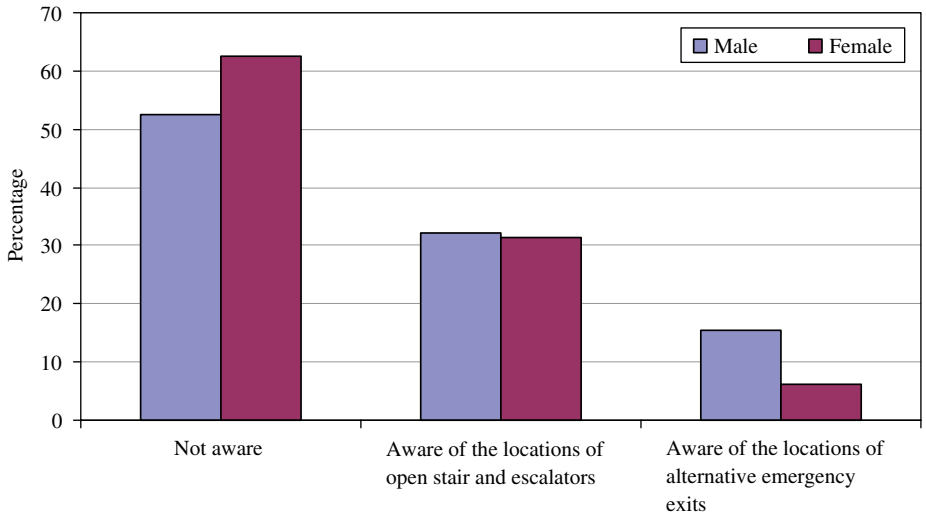


Fig. 2. Awareness of fire escape route.

Table 9
Observed frequency of awareness of fire escape route

Gender	Not aware	Open stair/ escalator	Open stair/ escalator and emergency exit	Total
Male	99	61	29	189
Female	132	66	13	211
Total	231	127	42	400

Table 10
Expected frequency of awareness of fire escape route

Gender	Not aware	Open stair/ escalator	Open stair/ escalator and emergency exit	Total
Male	109	60	20	189
Female	122	67	22	211
Total	231	127	42	400

Table 11
χ^2 and Cramer's V tests on gender dependence of awareness of escape route

χ^2 Test		Cramer's V
χ^2 (Obtained)	χ^2 (Critical)	
9.500	5.991	0.154

that the commuters' awareness of fire escape routes was independent of their gender. The results are presented in Table 11. The null hypothesis is rejected because the χ^2 (critical) is lower than the χ^2 (obtained). Therefore, the commuter's awareness of fire escape routes was dependent on their gender with males having better awareness. However, Cramer's V test revealed a very small value, and hence, weak association or correlation between gender and the awareness of fire escape route, contrary to the result of the χ^2 test. Seaman [17] highlighted the difficulties in interpreting Cramer's V. The proximity of Cramer's V to zero can be misleading because it "often corresponds to quite large proportional differences between groups." Seaman [17] further suggested that it was more reasonable to compare distributions than focusing solely on Cramer's V. He stressed that the latter should be a secondary index. Another reason for the weak association revealed by the Cramer's V test might be related to the sample size as discussed in the foregoing paragraph.

5. Results and discussions: walking speeds

Video surveillance and on-site observations were carried out at MRT stations, subways and pedestrian overhead bridges near MRT stations [15] to determine the walking speed of pedestrians. Video surveillance is one of the common methods used by researchers to collect pedestrian data [8]. Commuters were randomly selected and tacked in the video footage between landmarks of known distance. Also, a randomly selected number of commuters were physically tracked on site to determine their walking speeds. Only impeded walking speeds were taken into consideration for this project. Fruin's [7] criteria, as stated earlier in Section 2.3 of the current paper, were used to determine the congestion status.

The designs of the walkways and staircases under surveillance were compliant to the relevant standard of Singapore [18]. The dimensions are given in Tables 12 and 13, respectively.

Table 12
Dimensions of walkways

Walkway no.	Length (m)	Width (m)
1	64	4.8
2	49	2
3	9.6	3
4	7.2	2
5	6.3	3
6	3.5	1.2

Table 13
Dimensions of staircases

Stair no.	Width (m)	No. of steps	Rise (m)
1	4.8	15	0.16
2	2	13	0.17
3	2	12	0.17
4	2	15	0.17
5	2	16	0.175
6	1.8	12	0.15

5.1. Walking speed on walkways

The results of unimpeded walking speed on walkways are presented in Table 14 against age and gender groups. The impediment was determined by the pedestrian density threshold of 0.43 person/m^2 as per Fruin's LOS A and B [7]. When the pedestrian density is greater than this threshold, the pedestrian movement is considered to be impeded. The unimpeded average speed of adults was the fastest among all age groups. The unimpeded average walking speeds of female and male adults were 1.24 and 1.30 m/s, respectively. It was also found that the unimpeded average walking speeds of the female and male elderly were 1.04 and 1.05 m/s, respectively. Children could walk slightly faster than the elderly could and their average walking speed was 1.08 m/s. The average and median walking speeds of all age and gender groups exceeded the unimpeded speed of 1 m/s specified in SFSRTS [11].

The walking speeds of adults under a range of pedestrian density are shown in Fig. 3 as scattered points. Linear regression was conducted and the result is given in Table 17. From Fig. 3, it can be found that the regression line is slightly different from Fruin's [7] in general. Based on unimpeded walking speed for density between 0 and 0.43 person/m^2, the difference between the two regression lines is not more than 0.1 m/s. For pedestrian density more than 0.3 person/m^2, the regression line falls above that of Fruin's. The slightly faster movement speed obtained in the current study than that obtained by Fruin at higher pedestrian densities may be attributed to the smaller physiques of Singaporeans relative to that of Europeans and North Americans. At the same pedestrian density, smaller physiques of pedestrians means greater interspaces between them, hence allowing faster movement.

The measured pedestrian flow volumes are plotted against the parameter of space per pedestrian, γ, in Fig. 4. Similarly, regression approach was used to derive best curve fit and the result is included in Table 17. Because of faster walking speed by Singaporeans than North Americans at high pedestrian density, the pedestrian flow measured in the MRT stations in Singapore was higher than that obtained by Fruin [7] for low space per pedestrian.

5.2. Downstairs walking speed

The results of unimpeded downstairs walking speed are presented in Table 15. In this case, the threshold for impediment

Table 14
Unimpeded walking speeds on walkways ($\rho < 0.43$ person/m^2)

Age and gender group	Average (m/s)	Median (m/s)	s (m/s)	Probability of $v \leqslant 1$ m/s (%)	Number of samples
Children	1.08	1.06	0.23	36.79	283
Female elderly	1.04	1.04	0.21	42.89	114
Male elderly	1.05	1.08	0.20	40.64	104
Elderly	1.04	1.05	0.21	42.45	218
Female adult	1.24	1.20	0.14	4.32	163
Male adult	1.30	1.29	0.13	1.05	121
Adult	1.27	1.27	0.14	3.02	284
Total number of participants					785

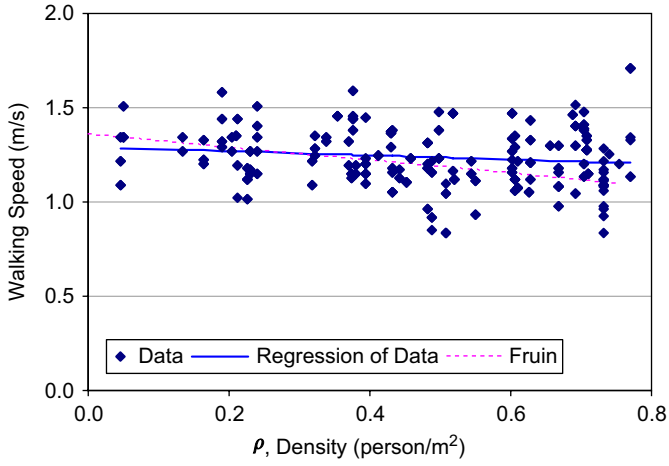


Fig. 3. Pedestrian speeds on walkways.

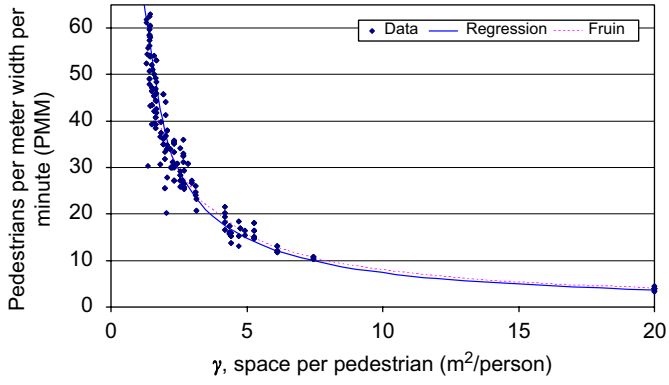


Fig. 4. Pedestrian flow volume on walkways.

was 0.72 person/m^2 as per Fruin's LOS A and B. The male adults were found to be the fastest walking group with average vertical downstairs walking speed of more than 0.42 m/s. The average vertical speed of adult was about 0.38 m/s. The average vertical walking speed of children was 0.31 m/s and was faster than that of elderly who had average walking speed of 0.28 m/s.

In general, the average and the median unimpeded vertical downstairs walking speeds of adult were found to be faster than average 0.3 m/s adopted by SFSRTS [11] and Fahy and Proulx [10]. Majority of the elderly did not walk faster than 0.3 m/s. The average walking speed of elderly was about 27% slower than that of adult. This was quite close to the one-third difference observed by Fruin [7]. From Fig. 5, the linear regression line is found to fall above Fruin's correlation when pedestrian density is more than 1.5 person/m^2.

The results presented in Fig. 6 show that the measured pedestrian downstairs flow volume match closely with Fruin's pedestrian researches except at crowd congestion levels lower than 0.9 m^2/person.

5.3. Upstairs walking speed

The results of unimpeded upstairs walking speed are presented in Table 16. In this case, the threshold for impediment was 0.72 person/m^2. Again, the male adults were found to be the fastest walking group with average vertical walking speed of more than 0.32 m/s. The average vertical speed of adult was 0.31 m/s. The average vertical walking speed of children was 0.3 m/s, which was faster than that of elderly who clocked 0.28 m/s. The average and the median vertical upstairs walking speeds of all age and gender groups were found faster than the average 0.25 m/s adopted by the SFSRTS [11].

As shown in Fig. 7, the linear regression line is found to fall above Fruin's line with pedestrian density more than 1.7 person/m^2. This means that under the measured congested conditions, the observed walking speeds of people were slightly faster than that of the studies by Fruin [7].

The results presented in Fig. 8 indicate that there were no significant differences between Fruin's correlations of pedestrian upstairs flow volume vs. space per person and the correlations derived from this study.

The correlation coefficients of the linear regressions for all walking speeds were very low (see Table 17). This indicates a very weak linear correlation between walking speed and pedestrian density. One of the reasons is that the pedestrian ages were not differentiated in the regression analyses. In addition, the walking speed is a highly random variable even within an age group. Therefore, strong linear fits to the data presented in Figs. 3, 5 and 7 are not expected. Other factors, such as trip purpose, temperature, traffic composition, etc., may also influence pedestrian walking speeds. However, these factors were not investigated in the current study. The correlations between pedestrian flow volume and space per person, on the other hand, have reasonably good correlation coefficient (see Table 17). This can be attributed to the fact that the pedestrian flow volume is based on the ensemble average speed and is less sensitive to the deviations in the individual walking speed.

6. Conclusions

The gender distributions of MRT commuters were found to be statistically significantly different from the general population distributions of the general population in Singapore. The

Table 15
Unimpeded downstairs walking speeds ($\rho < 0.72$ person/m^2)

Age and gender group	Average (m/s)	Median (m/s)	S (m/s)	Probability of $v \leqslant 0.3$ m/s (%)	Number of samples
Children	0.31	0.31	0.09	45.16	181
Female elderly	0.26	0.27	0.07	69.77	149
Male elderly	0.29	0.29	0.07	54.39	173
Elderly	0.28	0.28	0.06	62.27	322
Female adult	0.36	0.35	0.07	20.15	164
Male adult	0.42	0.39	0.11	13.14	128
Adult	0.38	0.37	0.09	18.17	272
Total number of participants					795

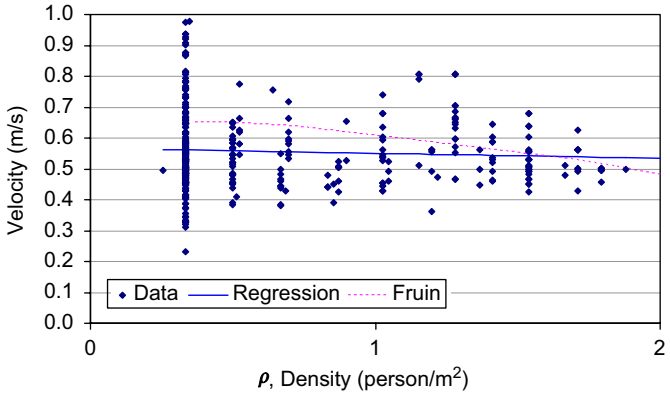


Fig. 5. Pedestrian downstairs movement speed.

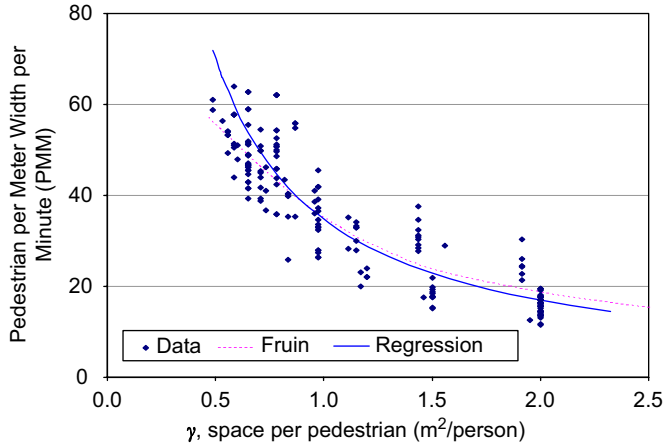


Fig. 6. Pedestrian downstairs flow volume.

percentages of elderly and children among commuters were below their respective percentages in general public. These differences may have influences on the occupant behaviour in general and in particular during fire and other emergencies in MRT stations and, therefore, should be considered in the design of MRT stations.

Based on the commuter survey, it was found that, regardless of gender, commuters intend more likely to respond to live announcement than to other types of warnings such as pre-recorded voice messages and fire alarm bells. The times required for the commuters in the MRT stations to leave the station upon hearing the live announcement could be different from that of the pre-recorded voice messages and other types of alarms. Thus, more in-depth studies on staffs' capabilities in handling different warning systems may be required in order to determine the effectiveness and efficiency of various systems.

A significantly large proportion of commuters was found unaware of or unfamiliar with the current fire escape route provisions in the MRT stations from the survey. This lack of awareness is likely to delay the commuters' evacuation during fire emergencies and, therefore, should be taken into consideration by the designers. More in-depth studies may be undertaken to determine the reasons of low awareness level and to explore the ways to improve the situation.

This study found that both male and female commuters respond similarly to different types of fire warning systems. However, there was an indication of gender difference in terms of familiarity with alternative emergency exit, notwithstanding the difference between the results of χ^2 and Cramer's V tests. Further study with larger sample size is required to verify the difference in the statistical analyses.

From the empirical observations, different average walking speeds were found for different gender and age groups. The results will make it possible to adopt different walking speeds of different population groups with the consideration of the population distributions. Such an approach would enable more reasonable estimate of movement time as compared with the generalised approach of adopting single walking speed.

Because of the variations in the walking speed with age and gender and the random nature of walking characteristics of pedestrians, correlations of the linear regression fits derived for walking speeds and pedestrian density were very weak. The regression curves of pedestrian flow volume appeared to have better correlations with the measured data than the pedestrian speed regression curves.

There were also slight differences between the regression lines derived from this study and previously published results by others. In particular, the pedestrian flow under high crowd congestion level obtained from the current study is higher than that obtained from the studies conducted in the USA. Thus, this study demonstrated a regional difference in the characteristics of pedestrian flow.

There may be interactions between different age groups, particularly between children and adult groups. The influence of children on adults' response and movement speed or vice versa deserves further investigation.

The observations, field surveys and walking speed measurements were conducted under normal or non-emergency conditions in this study. The results, particularly the survey results may not be accurate representations of the occupant behaviour during fire and other emergencies. They, nevertheless, provided indications of commuters' intentions and pre-incident mental attributes. As for the movement, it is envisaged that the computers' walking speed would be different under fire and other emergencies. Further studies are needed to verify and quantify the possible differences.

Table 16
Unimpeded upstairs walking speeds ($\rho < 0.72$ person/m^2)

Unimpeded	Average (m/s)	Median (m/s)	s (m/s)	Probability of $v \leqslant 0.25$ m/s (%)	Number of samples
Children	0.29	0.28	0.08	27.99	196
Female elderly	0.27	0.27	0.06	39.46	104
Male elderly	0.29	0.29	0.06	25.46	115
Elderly	0.28	0.28	0.06	32.38	219
Female adult	0.30	0.29	0.06	16.96	110
Male adult	0.32	0.30	0.08	19.31	118
Adult	0.31	0.30	0.07	19.00	228
Total number of participants					643

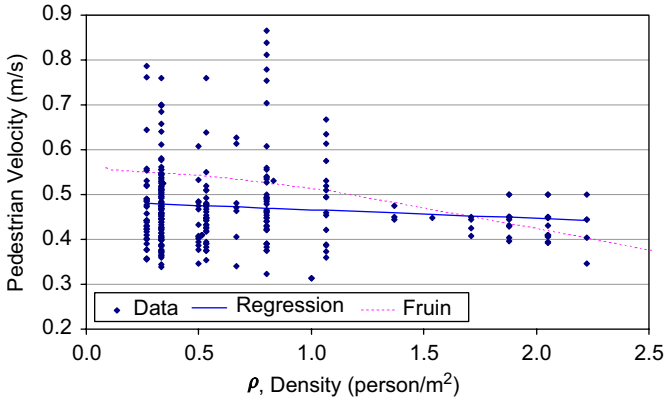


Fig. 7. Pedestrian upstairs movement speed.

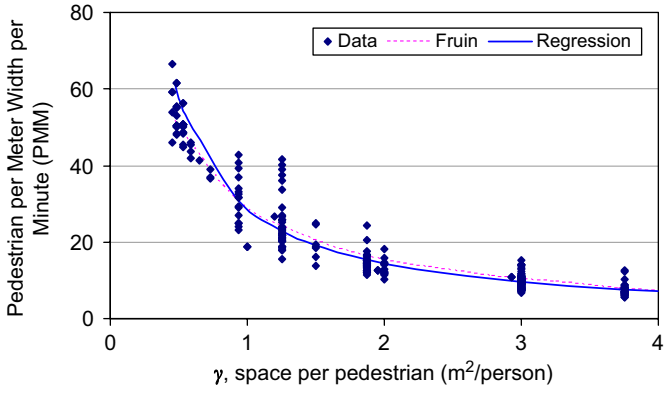


Fig. 8. Pedestrian upstairs flow volume.

Table 17
Walking speed and flow volume regression results

Walking direction	Parameter	Expression	r^2
Horizontal	Walking speed (m/s)	$v = 1.29 - 0.11\rho$	0.02
	Pedestrian flow volume (person/m min)	$F = -2.4 \times 10^{-4} + 73.85/\gamma$	0.89
Upstairs	Walking speed (m/s)	$v = 0.48 - 0.02\rho$	0.11
	Pedestrian flow volume (person/m min)	$F = 1.95 \times 10^{-4} + 28.77/\gamma$	0.92
Downstairs	Walking speed (m/s)	$v = 0.57 - 0.02\rho$	0.06
	Pedestrian flow volume (person/m min)	$F = -7.1 \times 10^{-4} + 35.43/\gamma$	0.90

The method used to obtain the results can be used and improved for future studies on commuter behaviour under unannounced fire drill conditions in the MRT stations and with more comprehensive video footage by mobile cameras and CCTV cameras.

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